

PHK Early Warning Dashboard

Technical Notes

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1 Methodology for Provincial PHK Analysis

1.1 Objective

- The objective of the analysis is to examine how macroeconomic pressures are associated with the monthly number of layoffs (PHK) across provinces, while accounting for differences in provincial economic and labor-market structures. The framework combines relatively slow-moving structural characteristics of provincial economies and labor markets with high-frequency macroeconomic indicators to capture two complementary dimensions of PHK risk: underlying structural vulnerability and short-term economic pressures. Lagged macroeconomic indicators are used to assess whether changes in macroeconomic conditions precede subsequent changes in PHK.
- This distinction is motivated by the literature on job destruction and regional economic restructuring. Aggregate economic shocks can generate substantial employment adjustment, but their effects are not necessarily uniform across sectors or regions. [Davis and Haltiwanger \(1999\)](#) show that oil-price and monetary shocks generate stronger short-run responses in job destruction than in job creation across most sectors, while the magnitude of these responses varies systematically with sectoral characteristics. This mechanism also has an important spatial dimension. Because economic activities and industries are unevenly distributed across locations, the geographic concentration of sectors can expose regions differently to structural change and sector-specific shocks. The OECD framework on industrial restructuring emphasizes this place-based nature of economic adjustment: regions with greater concentration in adversely affected activities may face stronger restructuring and employment pressures than regions with more diversified or less exposed economic structures ([OECD, 2026](#)).
- Building on this literature, the analysis conceptualizes PHK risk as arising from the combination of underlying structural vulnerability and evolving economic pressures. Provincial characteristics—including economic and sectoral structure, labor-market composition, wage conditions, and trade exposure—capture differences in the underlying vulnerability of provinces, while high-frequency macroeconomic indicators capture changing economic conditions that may trigger or amplify employment adjustment. The empirical framework therefore examines the association between macroeconomic pressures and PHK conditional on observable provincial structural characteristics. This distinction is particularly relevant for an early-warning framework: structural indicators identify *where* vulnerabilities are concentrated, while higher-frequency macroeconomic indicators provide signals of *when* those vulnerabilities may become more salient as economic conditions change.
- Conceptually, the framework can be summarized as follows:

$$\underbrace{\text{Structural characteristics}}_{\text{underlying vulnerability}} + \underbrace{\text{Macroeconomic pressures/shocks}}_{\text{dynamic triggers}} \longrightarrow \text{PHK risk.}$$

1.2 Econometric Approach

- The dependent variable is the monthly number of reported layoffs (PHK) by province, which is a non-negative count variable with a substantial number of zero observations.
- A Poisson regression with province fixed effects and calendar-month fixed effects is employed. The generic model specification is:

$$E(PHK_{it} | \mathbf{X}_{i,t-k}, \mathbf{S}_{it}) = \exp(\boldsymbol{\beta}'\mathbf{X}_{i,t-k} + \boldsymbol{\theta}'\mathbf{S}_{it} + \alpha_i + \delta_{q(t)}), \quad (1)$$

where:

- PHK_{it} denotes the number of reported layoffs in province i during month t ;
 - $\mathbf{X}_{i,t-k}$ denotes the macroeconomic pressure variable or vector of variables observed k months prior to month t . Depending on the specification, $\mathbf{X}_{i,t-k}$ may contain a single macroeconomic pressure indicator or multiple indicators included jointly. Macroeconomic indicators may be measured at the global, national, or provincial level; for indicators common across provinces, the province subscript i is not applicable;
 - $\mathbf{S}_{it}$ denotes a vector of provincial economic and labor-market structural characteristics included as control variables;
 - $\boldsymbol{\beta}$ denotes the coefficient or vector of coefficients associated with the lagged macroeconomic pressure variable(s);
 - $\boldsymbol{\theta}$ denotes the vector of coefficients associated with the provincial structural controls;
 - α_i denotes province fixed effects, which control for time-invariant differences across provinces;
 - $\delta_{q(t)}$ denotes calendar-quarter fixed effects (Q1–Q4), controlling for broad seasonal patterns that are common across provinces; and
 - $k \in \{1, 3, 6, 12\}$ denotes the lag horizon, measured in months.
- Robust standard errors are clustered at the provincial level to account for within-province serial correlation.
 - Two complementary specifications are estimated. First, each macroeconomic pressure indicator is entered separately, together with the common set of provincial structural controls, to assess its individual association with subsequent PHK. Second, all macroeconomic pressure indicators are entered jointly in single-lag specifications to assess their conditional associations while holding other observed macroeconomic pressures constant. Lag horizons of 1, 3, 6, and 12 months are considered separately.
 - Macroeconomic pressure indicators are measured at the global, national, or provincial level depending on data availability and the economic mechanism considered. The indicators enter the regressions at monthly frequency and are evaluated at lag horizons of 1, 3, 6, and 12 months.

- Provincial structural characteristics capture underlying differences in economic and labor-market conditions across provinces. Economic structure variables include indicators such as sectoral composition of GRDP, investment and trade exposure, GRDP per capita, and poverty. Labor-market structure variables include indicators such as unemployment, underemployment, employment composition, wages, and labor-force participation.
- Provincial structural variables are measured annually and assigned to the corresponding monthly observations within each year. These variables are included primarily as controls for differences in underlying provincial economic structure and labor-market vulnerability rather than interpreted as causal determinants of PHK.

1.3 Rationale for Model Selection

- **Poisson regression is appropriate for the count nature of the dependent variable.**

The dependent variable is the monthly number of reported layoffs (PHK) by province and is therefore a non-negative count variable that includes zero observations. A Poisson specification directly models the conditional mean of the count outcome and ensures non-negative fitted values. Importantly, consistency of the Poisson estimator does not require the dependent variable to follow an exact Poisson distribution, provided that the conditional mean is correctly specified. Robust inference is used to accommodate potential violations of the Poisson variance assumption ([Hausman et al., 1984](#); [Cameron and Trivedi, 2013](#)).

- **Province fixed effects account for persistent unobserved heterogeneity across provincial labour markets.**

Provinces differ systematically in industrial composition, economic structure, labour-market institutions, geography, and reporting practices. Some of these persistent characteristics may be correlated with the observed explanatory variables and with the incidence of reported PHK. Province fixed effects control for time-invariant provincial characteristics, allowing identification to rely on changes in explanatory variables within provinces over time ([Hausman et al., 1984](#); [Wooldridge, 2010](#)).

- **The panel structure allows macroeconomic pressures to be examined while accounting for differences in provincial exposure and vulnerability.**

The analysis combines monthly PHK observations with macroeconomic pressure indicators and provincial economic and labour-market characteristics. This framework is suitable for assessing whether changes in macroeconomic conditions precede subsequent changes in PHK while accounting for systematic differences in provincial structure. Lagged macroeconomic indicators at 1, 3, 6, and 12 months are examined to capture potentially different transmission horizons.

- **Calendar-month fixed effects account for recurring seasonal patterns without absorbing aggregate macroeconomic variation.**

Calendar-month fixed effects (January–December) are included to control for recurring seasonal patterns in reported PHK that may systematically differ across months of the year. Unlike a full

set of month-year fixed effects, calendar-month fixed effects preserve the time-series variation in macroeconomic indicators that is required to estimate their association with PHK.

- **Clustered standard errors account for within-province dependence over time.**

Standard errors are clustered at the province level, allowing for heteroskedasticity and arbitrary correlation of regression disturbances within provinces over time. This provides inference that is robust to serial dependence within provincial panels.

1.4 Variables

- Dependent Variable
 - Monthly Provincial PHK Flow (number of reported layoffs).
- Provincial Economic Structure (selected using PCA)
 - Share of Government Expenditure in GRDP (%)
 - Share of Investment in GRDP (%)
 - Share of Exports in GRDP (%)
 - Share of Imports in GRDP (%)
 - Share of Manufacturing in GRDP (%)
 - Share of Agriculture in GRDP (%)
 - Share of Mining in GRDP (%)
 - GRDP per Capita (log)
 - Poverty Rate (%)
- Provincial Labor Market Structure (selected using PCA)
 - Unemployment Rate (%)
 - Underemployment Share (%)
 - Labor Force Participation Rate (%)
 - Formal Labor Share (%)
 - Manufacturing Labor Share (%)
 - Agriculture Labor Share (%)
 - Full-Time Worker Share (%)
 - Minimum Wage (% yoy)
 - Average Wage (log)
- Macroeconomic Pressure
 - Global
 - * Brent Crude Oil Price (log)

- National
 - * BI Rate (%)
 - * IDR/USD Exchange Rate (log)
 - * Manufacturing PMI
 - * IHPB - Wholesale Price Index
- Provincial
 - * IHK - Consumer Price Index
 - * Non-Performing Loans (NPL)/Credit Ratio
 - * Export Value (log)
 - * Import Value (log)

1.5 Expected Relationships and Interpretation of Results

- The empirical analysis examines whether changes in macroeconomic pressures systematically precede changes in provincial PHK. In general, adverse macroeconomic conditions are expected to increase pressure on firms through weaker demand, higher financing costs, rising input costs, or deteriorating business conditions, which may subsequently lead firms to adjust employment. However, the expected direction and timing of these relationships may differ across indicators because macroeconomic shocks can operate through multiple and potentially offsetting channels.
- For example, higher interest rates may increase financing costs and weaken aggregate demand, potentially increasing subsequent PHK. Deteriorating manufacturing activity may similarly signal weaker production and labour demand. In contrast, the effects of inflation, exchange-rate movements, commodity prices, and international trade may be more heterogeneous. Exchange-rate depreciation, for instance, may increase costs for import-dependent firms while improving competitiveness for exporters. Similarly, higher producer or consumer prices may reflect either cost pressures or stronger aggregate demand. The empirical analysis therefore focuses not only on the direction of estimated associations, but also on their statistical significance, timing, and consistency across alternative lag horizons.
- The Poisson model is estimated in log-linear form. Accordingly, the estimated coefficient β_j represents the change in the log of the expected count of PHK associated with a one-unit change in explanatory variable X_j , holding other included variables constant. Because coefficients expressed in log-count terms are not always intuitive for policy interpretation, the results are also reported as *Incidence Rate Ratios* (IRRs), obtained by exponentiating the estimated coefficients:

$$IRR_j = \exp(\beta_j). \quad (2)$$

- The analysis places particular emphasis on whether estimated associations are statistically significant and consistent across the 1-, 3-, 6-, and 12-month lag horizons. Associations that appear

consistently across multiple lag horizons provide stronger descriptive evidence of a systematic relationship with subsequent PHK, whereas isolated or sign-changing estimates are interpreted more cautiously. These relationships are interpreted as conditional associations and potential leading relationships, rather than as causal effects or validated forecasting relationships.

- The magnitude of an IRR must, however, be interpreted according to the unit and transformation of the corresponding explanatory variable. A one-unit change has different economic meanings for variables measured in percentage points, index points, or logarithms. For variables expressed in logarithms, the reported IRR corresponds to a one-unit change in the log variable and should therefore not generally be interpreted directly as the effect of a one-percent change. For policy interpretation, economically meaningful changes—such as a one-percentage-point increase in an interest rate or a specified percentage change in a logged variable—should be used when translating estimated coefficients into changes in expected PHK.

2 Results

2.1 Macroeconomic Pressure and Leading Indicator Analysis

Table 1: Direction of Significant Associations between Lagged Macroeconomic Indicators and PHK

Variable	Lag 1	Lag 3	Lag 6	Lag 12
price_brent_yoy			+	
pmi_manuf	-			
bi_rate	+	+	+	+
fx_volatility		-	+	-
ihpb_index			+	+
car_sales_yoy			+	
log_export	-	-		

Notes: Each cell summarizes the estimated association from a separate Poisson regression of monthly provincial PHK on the indicated lagged macroeconomic pressure variable, controlling for provincial economic and labor-market structure, province fixed effects, and calendar-quarter fixed effects. '+' ('-') denotes a positive (negative) association statistically significant at the 10% level or better. Blank cells denote estimates that are not statistically significant at the 10% level.

Table 2: Incidence Rate Ratios (IRRs) from Poisson Regressions

Variable	Lag 1	Lag 3	Lag 6	Lag 12
price_brent_yoy	1.01	1.00	1.01**	1.01
pmi_manuf	0.94***	0.99	1.01	0.99
bi_rate	1.75**	2.22***	1.87***	1.35***
fx_volatility	1.00	1.00***	1.00***	1.00**
ihpb_index	1.07	1.06	1.15**	1.09**
car_sales_yoy	1.00	1.00	1.01**	0.99
log_export	0.64*	0.40**	1.12	0.98

Notes: Each cell reports the incidence rate ratio (IRR) from a separate Poisson regression of monthly provincial PHK on the indicated lagged macroeconomic pressure variable, controlling for provincial economic and labor-market structure, province fixed effects, and calendar-quarter fixed effects. IRRs are obtained by exponentiating the estimated Poisson coefficients. An IRR above (below) one indicates a positive (negative) association with expected PHK. The magnitude of the IRR should be interpreted according to the unit and transformation of each explanatory variable. Standard errors are clustered at the province level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

2.2 Poisson Regression: Single-Lag Specifications

Table 3: Poisson Regression Estimates of PHK on Macroeconomic Pressures

	(1)	(2)	(3)	(4)
	Lag 1	Lag 3	Lag 6	Lag 12
Macroeconomic Pressure (with lag)				
L_price_brent_yoy	0.000 (0.011)	-0.012* (0.006)	-0.005 (0.014)	0.000 (0.010)
L_pmi_manuf	-0.038 (0.031)	-0.056 (0.069)	0.052 (0.060)	-0.067 (0.126)
L_bi_rate	0.713** (0.246)	1.404*** (0.322)	1.540* (0.708)	0.224 (1.123)
L_fx_volatility	0.001 (0.002)	-0.004* (0.002)	0.002* (0.001)	-0.004*** (0.001)
L_ihpb_index	0.209*** (0.045)	0.135 (0.085)	0.072 (0.082)	0.093 (0.083)
L_car_sales_yoy	0.003 (0.009)	-0.004 (0.005)	0.012* (0.006)	-0.004 (0.010)
L_log_export	-0.398 (0.221)	-0.666 (0.501)	0.158 (0.325)	0.033 (0.253)
Provincial Economic Structure				
gov_pdrb_share	-0.199 (0.189)	-0.284 (0.228)	-0.271 (0.188)	0.062 (0.424)
export_pdrb_share	0.280 (0.169)	0.343* (0.169)	0.132 (0.139)	-0.412 (0.436)
import_pdrb_share	0.542** (0.193)	0.352 (0.201)	0.349 (0.245)	0.256 (0.528)
manuf_pdrb_share	0.154 (0.176)	0.029 (0.177)	-0.000 (0.158)	0.329 (0.198)
agri_pdrb_share	1.141 (0.590)	0.906 (0.546)	0.820 (0.688)	0.116 (0.691)
Provincial Labor Market Structure				
formal_lab_share	-0.092 (0.090)	-0.120 (0.084)	-0.063 (0.082)	0.074 (0.156)
manuf_emp_share	0.027 (0.123)	0.006 (0.117)	-0.065 (0.141)	-0.625* (0.286)
full_time_share	-0.113 (0.092)	-0.121 (0.078)	-0.076 (0.103)	-0.042 (0.108)
underemployment_share	-0.139 (0.114)	-0.101 (0.134)	0.164 (0.194)	1.093*** (0.305)
log_avg_wage	3.051 (2.535)	3.250 (2.096)	2.316 (2.126)	0.949 (2.480)
Observations	1211	1141	1036	827
Monthly PHK Mean	192	194	201	205
Pseudo R ²	0.599	0.612	0.630	0.643
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of PHK by province. Columns (1)–(4) report separate Poisson specifications in which all macroeconomic pressure indicators enter at lags of 1, 3, 6, and 12 months, respectively. Provincial economic and labor-market structural controls enter contemporaneously. All specifications include province and calendar-quarter fixed effects. Standard errors clustered at the province level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

Table 4: Incidence Rate Ratios (IRR) from Poisson Regressions of PHK on Macroeconomic Pressures

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Macroeconomic Pressure (with lag)				
L_price_brent_yoy	1.00 (0.01)	0.99* (0.01)	1.00 (0.01)	1.00 (0.01)
L_pmi_manuf	0.96 (0.03)	0.95 (0.06)	1.05 (0.06)	0.93 (0.12)
L_bi_rate	2.04** (0.50)	4.07*** (1.31)	4.67* (3.30)	1.25 (1.41)
L_fx_volatility	1.00 (0.00)	1.00* (0.00)	1.00* (0.00)	1.00*** (0.00)
L_ihpb_index	1.23*** (0.06)	1.14 (0.10)	1.07 (0.09)	1.10 (0.09)
L_car_sales_yoy	1.00 (0.01)	1.00 (0.00)	1.01* (0.01)	1.00 (0.01)
L_log_export	0.67 (0.15)	0.51 (0.26)	1.17 (0.38)	1.03 (0.26)
Provincial Economic Structure				
gov_pdrb_share	0.82 (0.15)	0.75 (0.17)	0.76 (0.14)	1.06 (0.45)
export_pdrb_share	1.32 (0.22)	1.41* (0.24)	1.14 (0.16)	0.66 (0.29)
import_pdrb_share	1.72** (0.33)	1.42 (0.29)	1.42 (0.35)	1.29 (0.68)
manuf_pdrb_share	1.17 (0.20)	1.03 (0.18)	1.00 (0.16)	1.39 (0.27)
agri_pdrb_share	3.13 (1.85)	2.47 (1.35)	2.27 (1.56)	1.12 (0.78)
Provincial Labor Market Structure				
formal_lab_share	0.91 (0.08)	0.89 (0.07)	0.94 (0.08)	1.08 (0.17)
manuf_emp_share	1.03 (0.13)	1.01 (0.12)	0.94 (0.13)	0.54* (0.15)
full_time_share	0.89 (0.08)	0.89 (0.07)	0.93 (0.10)	0.96 (0.10)
underemployment_share	0.87 (0.10)	0.90 (0.12)	1.18 (0.23)	2.98*** (0.91)
log_avg_wage	21.14 (53.59)	25.79 (54.05)	10.13 (21.53)	2.58 (6.40)
Observations	1211	1141	1036	827
Monthly PHK Mean	192	194	201	205
Pseudo R ²	0.599	0.612	0.630	0.643
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: Entries report incidence rate ratios (IRRs), obtained by exponentiating the corresponding Poisson coefficients. Columns (1)–(4) jointly include all macroeconomic pressure indicators at lags of 1, 3, 6, and 12 months, respectively, together with contemporaneous provincial economic and labor-market structural controls, province fixed effects, and calendar-quarter fixed effects. An IRR above (below) one indicates a positive (negative) conditional association with expected PHK. The magnitude of each IRR should be interpreted according to the unit and transformation of the corresponding explanatory variable. Standard errors clustered at the province level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

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